

Coupled simulations with preCICE

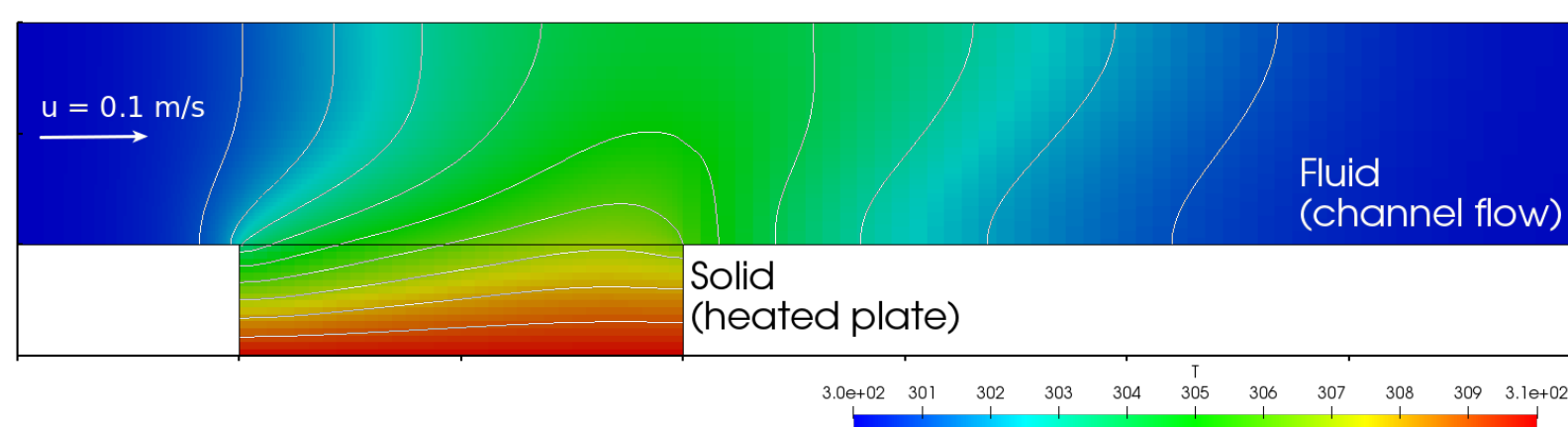
and with FAIR community contributions

Plain language summary

Single computer models are often not enough to describe some phenomena. Examples are the flow inside your pulsating heart, the ice-sheet melting in the arctic, or a material with tangible behavior defined by phenomena at the smallest scales. The software library preCICE allows programmers and end users to combine existing models with minimal effort, producing accurate results and running on a laptop or a supercomputer. Recent improvements have made the software easier to use, faster, more accurate, and fitting for new types of coupling. We now need to guide the rapidly growing community with producing and sharing quality contributions, and to provide tools to make building complex simulation scenarios easier.

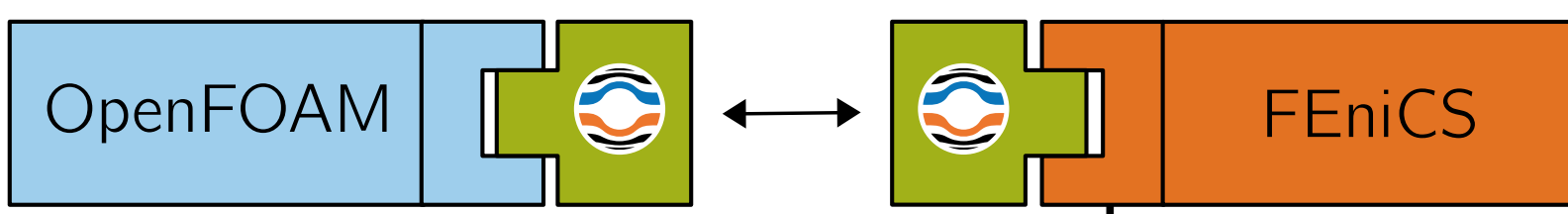
A partitioned simulation example

Flow over a heated plate: A preCICE tutorial. Choose your "hot solid" code among FEniCS, Nutils, CalculiX, OpenFOAM, and more. Simulate the flow with OpenFOAM or your own code.



Coupling via preCICE^[1,2]

Add calls to the preCICE library into each code, or use provided adapters. Start each code in a separate process (e.g., two terminals).



```
import preCICE
v = preCICE.set_mesh_vertices(positions)
preCICE.initialize()
while preCICE.is_coupling_ongoing():
    solve_as_usual()
    preCICE.write_data("temperature", v)
    preCICE.advance()
    preCICE.read_data("heat-flux", v)
```

Communication: TCP/IP sockets, MPI ports

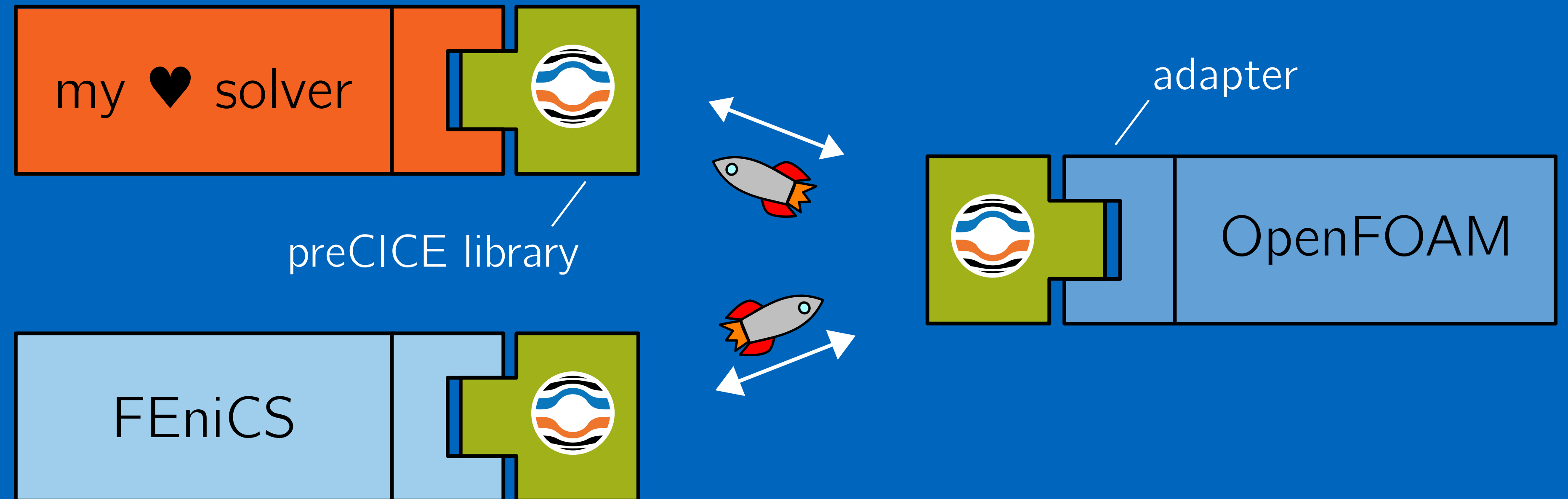
Mapping: kernel methods (RBF), projection methods

Coupling: Anderson acceleration (IQN), Aitken, ...

Time interpolation: waveform iteration

New in preCICE v3 / v2404^[3]

- Multirate and higher-order time stepping
- Much faster RBF mapping (partition of unity)
- Geometric multiscale mapping, micro-macro coupling
- More useful profiling tools, system tests
- Simpler API and configuration, default values

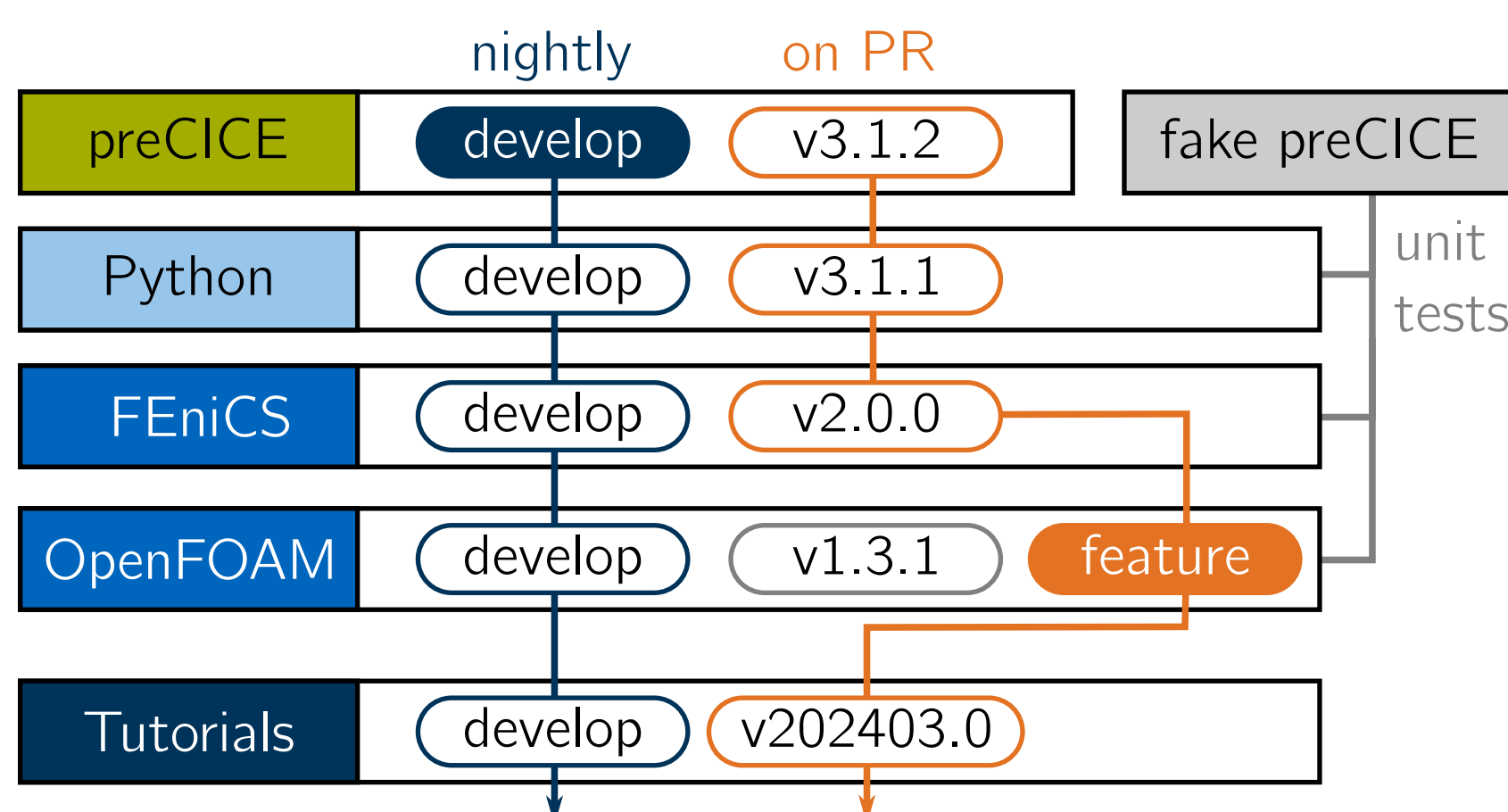


The preCICE ecosystem

A citable ecosystem of components

citable preCICE Distribution (on DaRUS ^[3])			
Website + Documentation	Nix pkgs	VM	
Tutorials			
OpenFOAM	CalculiX	FEniCS	deal.II
SU2	DUNE	DuMuX	MBDyn
Python	Julia	Matlab	Rust
preCICE (C++ core, C & Fortran bindings)			

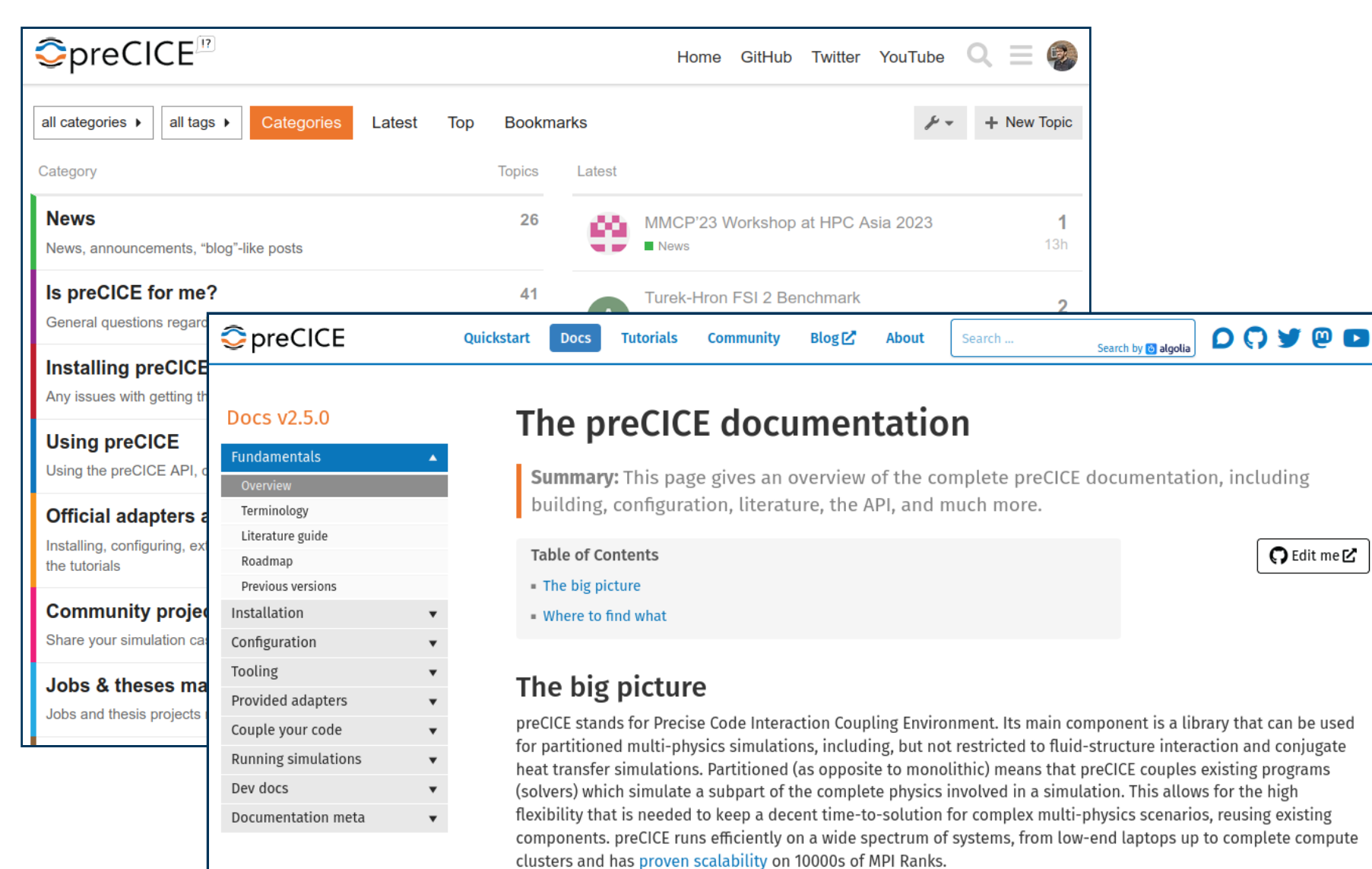
Testing the ecosystem^[4]



Can we release preCICE v3? Can we merge the PR?

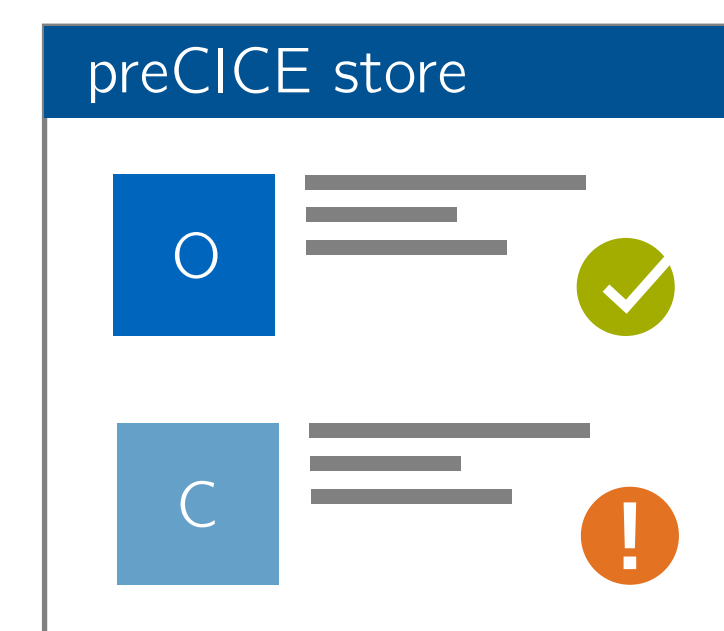
Forming a community^[2]

- Go to the users (domain conferences).
- Answer questions by editing the documentation.
- Train the users and get feedback (workshops).



The preECO project^[5]

Community-contributed components



We need a standard!

- ☐ Metadata, citation guidelines
- ☐ Tests available
- ☐ Cases validated
- ☐ Best practices

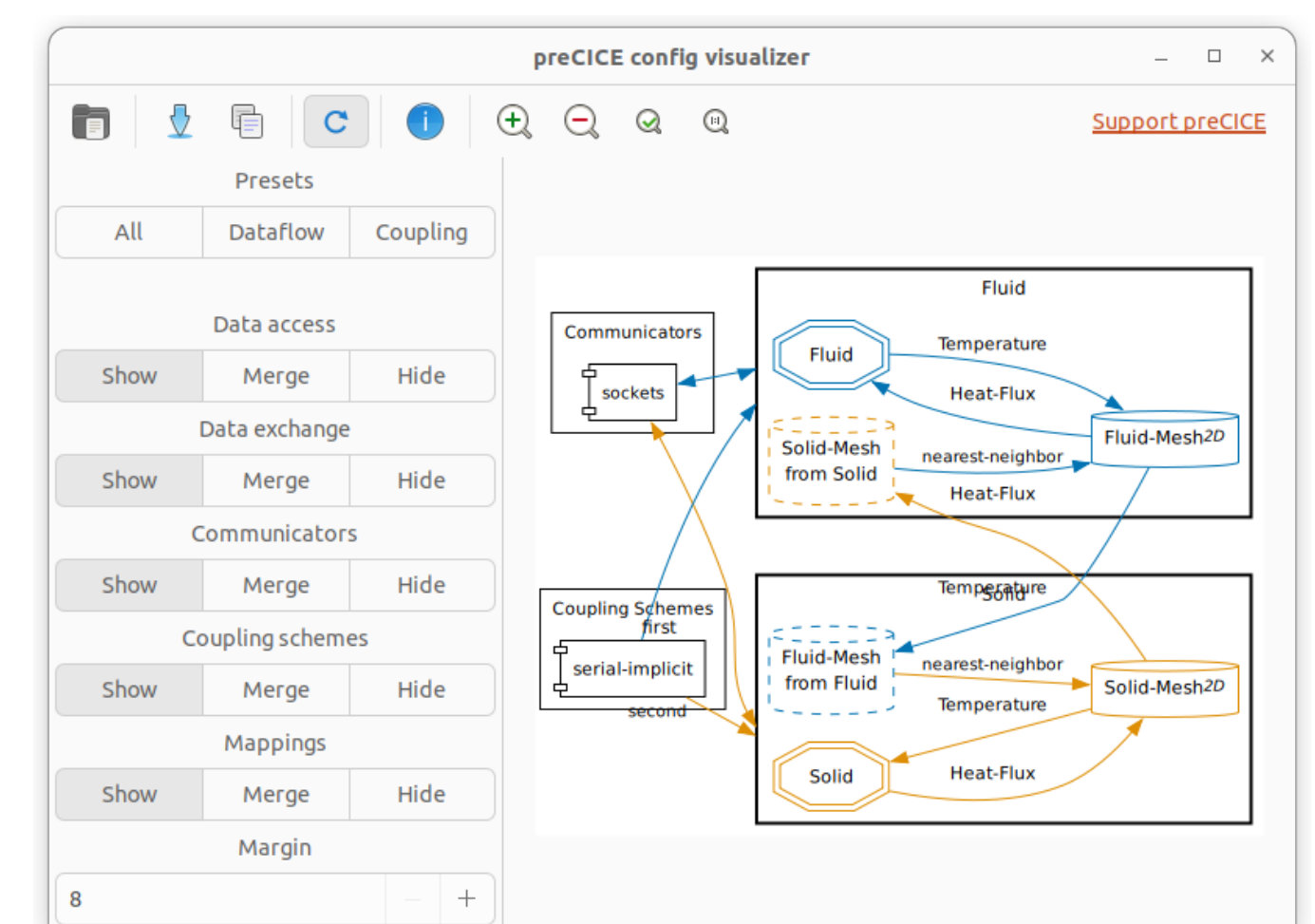
preCICE conforming Gold level

preCICE conforming Silver level

preCICE conforming Bronze level

Configuration and testing tools

precice-tools check precice-config.xml
precice-tools generate topology.yaml



Training and supporting the community

- Course modules: Basic, Tooling, Coupling, Mapping
- Offered at preCICE workshops, conferences, or on-demand
- New module: Best preCICE practices
- Academic funding + support licenses (apply in proposals)

Timeline

preCICE Workshop 2024/2025/2026:

- Develop the standards together with the community
- Develop and get feedback on the configuration tools
- Apply on the current adapters and tutorials
- Develop and execute a component for the training course

The preCICE v3 coupling library and the emerging preCICE ecosystem

The preCICE team (2024):

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<https://go.tum.de/595541>



[1] H.-J. Bungartz, F. Lindner, B. Gatzhammer, M. Mehl, K. Scheufele, A. Shukae, B. Uekermann (2016). preCICE – A fully parallel library for multi-physics surface coupling. In Comput. & Fluids, 141, 250–258. <https://doi.org/10.1016/j.compfluid.2016.04.003>

[2] G. Chourdakis, K. Davis, B. Rodenberg, M. Schulte, F. Simonis, B. Uekermann, et al., preCICE v2: A sustainable and user-friendly coupling library [version 2; peer review: 2 approved] Open Res Europe 2022, 2:51, <https://doi.org/10.12688/openreseurope.14445.2>

[3] J. Chen, G. Chourdakis, I. Desai, C. Homs-Pons, B. Rodenberg, D. Schneider, F. Simonis, B. Uekermann, K. Davis, A. Jaust, M. Kelm, N. Kotarsky, H. Kschidock, D. Mishra, M. Mülhaußer, T. P. Schrader, M. Schulte, V. Seitz, J. Signorelli, G. van Zwieten, N. Vinnitschenko, T. Vladimirova, L. Willeke, E. Zonta (2024) preCICE Distribution Version v2204.0 <https://doi.org/10.18419/darus-4167>, DaRUS, V1.

[4] G. Chourdakis, V. Seitz, B. Uekermann (2024). System Regression Tests for the preCICE Coupling Ecosystem. In proceedings of the derSE24, Würzburg. ECEASST (in review)

[5] M. Schulte, B. Uekermann (2023). A Community-Driven Ecosystem of Adapters and Application Cases for the Coupling Library preCICE. Project description, Deutsche Forschungsgemeinschaft. GEPRIS database: <https://gepris.dfg.de/gepris/projekt/528693298?language=en>